Question 1:

A loop structure repeats a set of code or instructions while a condition is true, or for a specific amount of times.

Question 2:

A while statement checks the condition before starting the code so it might never run at all. A do- while statement checks the conditions after the code has been run, so it runs at least once

Question 3:

Question 4:

a) A loop that never ends because no conditions were set to be false.
b) A logical error in the loop conditions or forgetting to update the loop control variable
c) Overflow occurs when a variable is given a value that exceeds the max or min value it can store.

Question 5: 60

Question 6: any value less than 120

Question7: counter: keeps track of how many times something occurs.

Uses: counting number of students and counting the amount of times a loop repeats

Accumulator: adds values together to maintain a running total

Uses: Calculating the sum of numbers and calculating the total cost of items.

Question 8: int sum = 0;

For (int num =3; num <=10; i++)

Question 9: whether the number of repetitions is known, and whether or not the loop must execute at least once

Question 11:

a) 10
b) “my “
c) “my string.”
d) “MY STRING.”
e) “my string.”